

1. Define wilting point. Explain crop-water requirements for crops: paddy, wheat, maize and potato grown in the Mahabharat region of Nepal.

Answer:

Plant roots are able to extract water from a soil matrix, which is saturated up to the field capacity. However, as the water extraction proceeds, the moisture content diminishes, and the negative pressure increases, at one point, the plant cannot extract further water and thus wilts. This point is known as wilting point.

There are two stages of wilting point. They are:

i. Temporary wilting point:

This point denotes the soil water content at which the plant wilts at day time but recovers during night or when the water is added to the soil.

ii. Permanent wilting point:

This point denotes the soil water content at which the plant wilts and fail to regain life even after addition of water to the soil.

Crop Water Requirement (CWR) is defined as the total quantity of water required by a crop from the time it is sown to the time it is harvested. The Mahabharat region of Nepal extends from the altitude of 1500-3300 m above the mean sea level, has a subtropical to temperate climate, with most rainfall during the monsoon (June–September). Hence, Supplementary irrigation has to be provided, especially during the dry season.

For paddy crop requires large amount of water, the field is completely flooded by rainwater and if required by irrigation facilities. So, the plant is grown in monsoon season from June-October. It requires approximately 1200-1600 mm of water.

Wheat is grown in November-April season, as it can grow in less amount of water. It requires 450-650 mm of water in the field.

Maize is grown primarily in summer season, which can be grown in March-July or July-November. It requires 500-800 mm depth of water for its growth.

Potatoes can be grown across different seasons in Nepal depending upon the topographical regions and the altitude. It requires approximately 400-600 mm depth of water.

2. What are the various methods of applying irrigation water to land? It is said that user's participation is a must from planning state of an irrigation project for its success. Explain? How do you proceed for the preparation and development of an irrigation project? Also mention the various types of survey to be conducted.

Answer:

Irrigation is an artificial application of water to agricultural land in the required quantity and at the required time to supplement natural rainfall for optimum crop growth and yield.

There are several methods of irrigation adopted based on the suitability of local condition such as soil characteristics, topographic conditions, crop rotation, available water flow and others. Few methods of applying water in irrigation fields are briefly described below:

i. Surface irrigation:

- Surface irrigation is the most common type of the irrigation where the water is applied on the field in varied quantities and at different times.

- In this method, stream of water is diverted from the head of a field into the furrows or borders and allowing it to flow downward.
- Suitability:
 - Surface irrigation is most suitable for the land with the gentle slopes like terai region of Nepal.
 - This method is suitable for row and field crops.

Surface irrigation is further classified into:

- a. Flooding method: entire land surface is covered with water and allowed to stand on the field.
- b. Contour farming: adapted to hilly areas with steep slopes and quickly falling contours.
- c. Furrow irrigation: common method of irrigation where water is applied in small streams between the rows of the crops, grown on the ridges or furrows.



ii. **Sprinkler irrigation:**

- Sprinkler irrigation is an improvement over conventional surface irrigation method.
- This method simulates natural rainfall to spread water in the form of rain uniformly over the land surface as needed at uniform pattern and at less rate than the infiltration rate of the soil to avoid the surface runoff.
- Suitability:
 - Sprinkler irrigation is highly suitable for sandy soils, rolling and uneven terrain.
 - It is suitable for closely spaced crops like cereals and fodders.



iii. **Drip irrigation:**

- Drip irrigation is the method of applying water directly to the plants through a number of low flow rate outlet called emitter or drippers, placed at short interval along with small tubing.
- The network of drippers supply water and fertilizer to the plant roots at the rate of 1 to 2 liter per hour.
- Suitability:
 - This method of irrigation is highly suitable for arid regions and sloppy terrain where the water is scarce.
 - The method is ideal for row crops, orchard and vineyards.



User's participation is defined as a process where beneficiaries, community organizations, and other stakeholders are actively involved in every stage of a project, including planning, design, implementation, operation, maintenance, and monitoring. User's participation is a must from planning state of an irrigation project for its success because of the following:

- i. Participation at the planning stage ensures that the project addresses the actual needs and priorities of the community.
- ii. When users are involved in planning, project sites and canal alignments are selected based on community consensus and mutual agreement which minimizes future conflicts over land acquisition.
- iii. Early involvement of community users fosters a deep sense of ownership among them which results in being more likely to contribute their own labor and resources for its protection and long-term sustainability.
- iv. Local farmers possess significant indigenous knowledge regarding the local terrain, soil characteristics, and river behavior. Integrating this knowledge during the planning and design stages leads to more pragmatic, adapted, and cost-effective technological solutions.
- v. Planning with the community allows for the establishment of local rules for equitable water distribution and creates internal mechanisms for social cooperation and conflict resolution regarding water rights.

The preparation and development of an irrigation project can be proceeded in following steps:

The step-by-step logical design process for irrigation development of a medium scale irrigation project are listed below:

- i. Reconnaissance survey:
 - Identify the project area and water source.
 - Assess topography, climate, and preliminary feasibility.
 - Identify possible canal alignment and command area.
- ii. Feasibility study:
 - Assess water availability (surface or groundwater).
 - Estimate irrigation demand and command area.
 - Conduct preliminary economic, technical, social, and environmental analyses.
 - Select the most suitable project alternative.
- iii. Detailed survey:
 - Topographic survey.
 - Hydrological and meteorological investigations.
 - Geological and geotechnical investigations.
 - Soil survey and land capability assessment.
 - Command area survey.
- iv. Hydraulic design:

- Design headworks (weir/barrage/intake).
- Design main canal, branch canals, distributaries, and minors.
- Design canal sections, slopes, and velocities.
- Design cross-drainage works, falls, regulators, escapes, and outlets.

v. Structural design:

- Design foundations and hydraulic structures.
- Check stability against uplift, sliding, overturning, and seepage.
- Prepare structural drawings and reinforcement details.

vi. Environmental assessment:

- Conduct IEE/EIA as required.
- Prepare environmental management and mitigation plans.
- Address land acquisition, resettlement, and social impacts.

vii. Preparation of Detailed Project Report:

- Prepare design drawings.
- Estimate quantities and project cost.
- Prepare construction specifications and implementation schedule.

viii. Construction planning and implementation:

- Procure contractors and materials.
- Construct headworks, canals, and associated structures.
- Supervise quality control and project progress.

The various types of surveys to be conducted for irrigation projects are:

- i. Reconnaissance survey
- ii. Topographical survey
- iii. Geological and geotechnical survey
- iv. Hydrological survey
- v. Cadastral survey
- vi. Environmental survey etc.

3. **a. Write short notes on the following:**

i. Free board in canals ii. Berm of canal

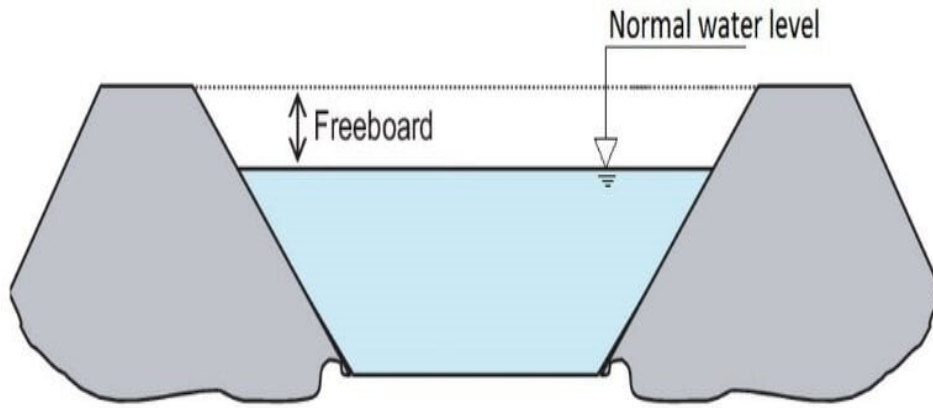
b. What do you understand by fall in a canal? Why is it necessary? How do you select in its location?

Answer:

i. Free board in canals:

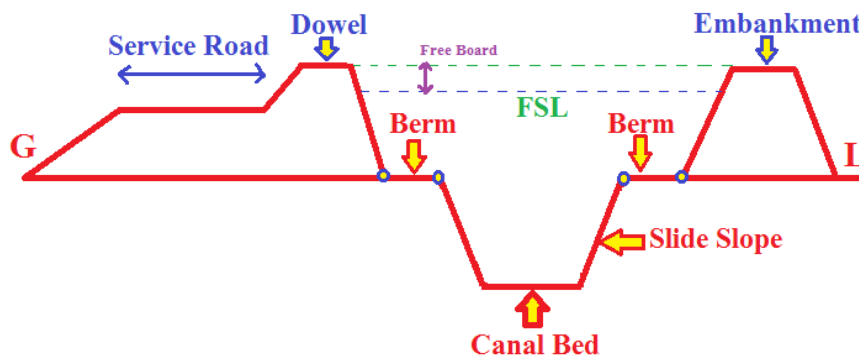
- Free board in canal is the vertical distance from the water surface at full supply level to the top of the bank.
- Freeboard provides the margin of safety against overtopping of the banks due to the sudden rise in the water surface of a channel on account of improper operation of gates at the head regulator.
- The excessive growth of vegetation or accumulation of sediment deposits may also result in gradual rise of water surface levels above the design levels.
- Design of channels should specify adequate freeboards to prevent overtopping of banks during the sudden rise in water surfaces.
- Adequate freeboard would depend on dimensions of the flow section, flow condition, bank materials, method of construction of banks and resulting damage due to failure of banks.
- Freeboard in unlined channels vary from about 0.3 m in small channels to about 2 m in large channels.
- For channels of intermediate size, freeboard is sometimes estimated by adding 0.3 m to one quarter of the flow depth.

Freeboard in a canal



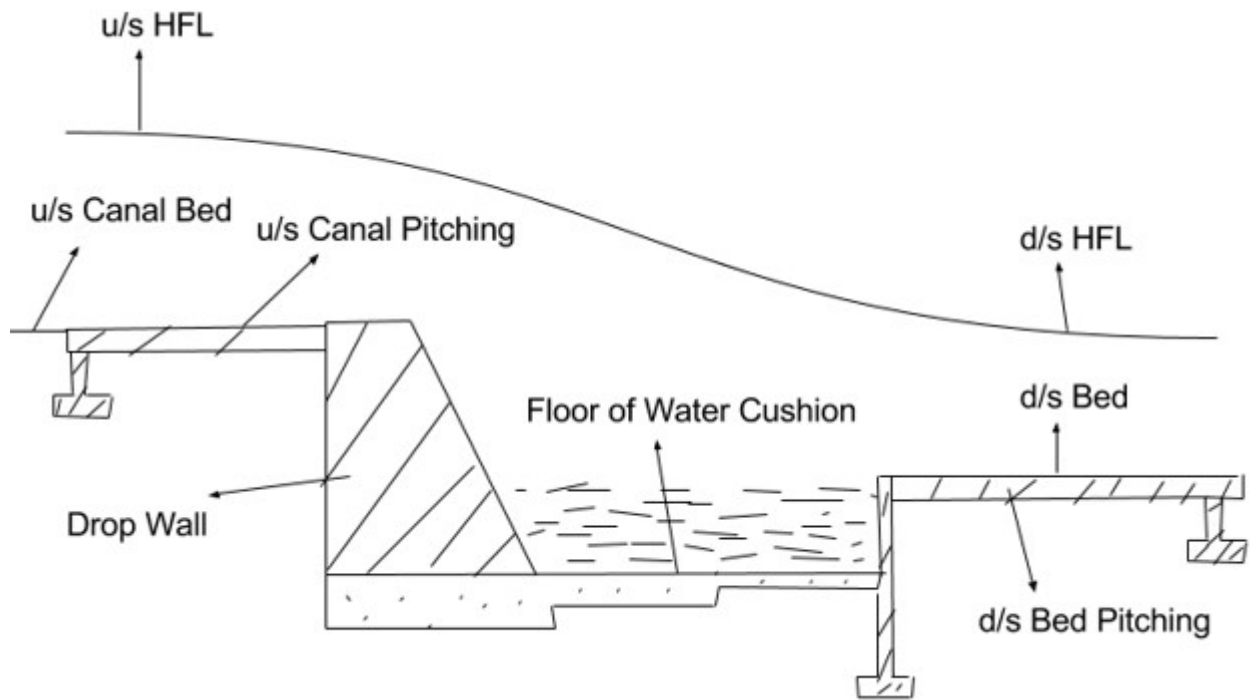
ii. Berm of canals:

- A berm is a narrow horizontal strip of land between the inner toe of the bank and the top edge of cutting.
- Berms between the water section and inner bank slopes are required along the channels where bank materials are susceptible to sloughing.
- Berms slope towards the water section to facilitate the drainage.
- An impervious lining is formed on the banks, as a result of silting on the berm which helps in reduction of seepage losses.
- Berms also help in the protection of banks against the breaches and eroding action of water waves.
- It breaks the flow of rainwater down the bank slope and prevents guttering.



b.

- A drop or fall in the canal is a regulating structure which lowers the water level along its course.
- The slope of a canal is usually milder than the terrain slope as a result of which the canal in a cutting at its headwork will soon outstrip the ground surface.
- To avoid the excessive infilling the bed level of the downstream canal is lowered, the two reaches being connected by a suitable drop structure. This is why the fall is necessary.
- Location of fall can be selected where:
 - i. The natural ground slope exceeds the designed canal bed slope.
 - ii. The fillings and cutting of the canal are balanced as much as possible.
 - iii. Stable and firm foundation soil can be found.
 - iv. Sufficient energy dissipation arrangements can be provided at the downstream of the fall.
 - v. Construction and future maintenance are convenient and economical.



4. **State Manning's formula for canal design. Define its various component. What is the effect of change in manning "N" in the flow of canals?**

Answer:

Manning's formula for canal design is given by:

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

Since,

$$Q = AV$$

Substitute the Manning velocity:

$$Q = \frac{A}{n} R^{2/3} S^{1/2}$$

Where:

Q = Discharge (m³/s)

A = Cross-sectional area (m²)

R = Hydraulic radius (m)

S = Bed slope

n = Manning's roughness coefficient

Velocity equation:

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

Typical values:

- Smooth concrete canal: 0.011–0.015
- Brick masonry: 0.015–0.017

- Earth canal (good condition): 0.020–0.030
- Rough natural channel: 0.030–0.050 or more

From Manning's equation,

$$V \propto \frac{1}{n}$$

Thus, velocity and discharge are inversely proportional to Manning's roughness coefficient.

The increase in the value of n , results in rough canal surface which reduces the flow velocity and canal discharge. Also, sediment deposition increases due to lower velocity.

The reduction in the value of n , results in smooth canal surface which increases flow velocity and canal discharge. Excessively high velocity may also cause scouring and erosion in unlined canal.

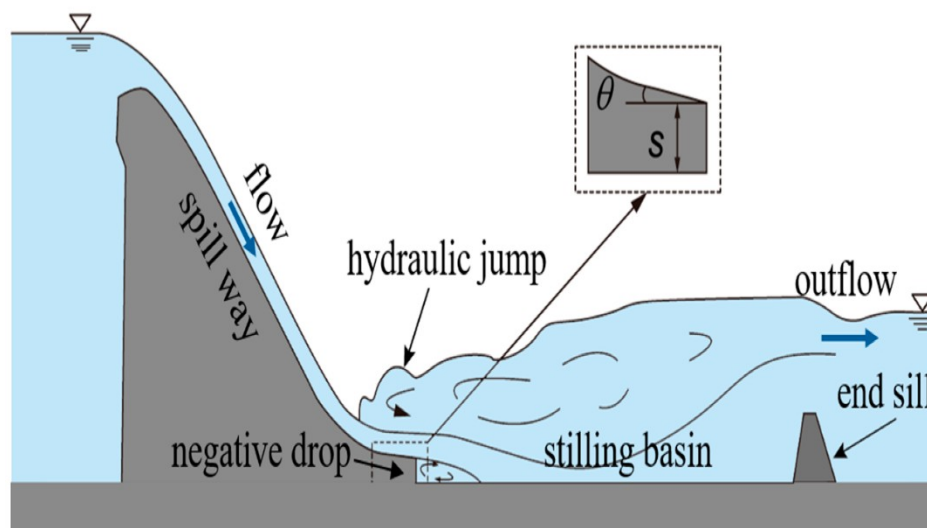
5. **What is hydraulic jump? How it occurs and what is its use in hydraulic structures. What is the importance of water management work in distribution system of canal?**

Answer:

- When the depth of flow changes rapidly from a low stage to a high stage, there is abrupt rise of water surface. This local phenomenon is known as hydraulic jump.
- It occurs when the flow changes from supercritical to subcritical.
- Hydraulic jump converts kinetic energy of stream flowing rapidly into potential energy.
- It occurs frequently in a channel below a regulating sluice, at toe of spillway and when the channel bed slope suddenly changes from the steeper to flat slope or where the velocity is high.
- Hydraulic jump forms eddy turbulences in water which causes dissipation of energy.

It is necessary to be formed in hydraulic structures which is explained in the points below:

- For the dissipation of excess energy of flowing water over spillway and other hydraulic structure.
- For preventing from possible erosion of downstream floor at structure.
- For increasing the weight of apron floor of hydraulic structure due to increased flow depth.
- For mixing chemicals in water by the help of turbulence created.
- For raising the water level in the channel for irrigation.
- For increasing the discharge through a sluice by holding back tail water.



For a stable and well-balanced hydraulic jump, the Froude number (Fr) before the jump should be greater than 1 i.e., the flow must be supercritical.

The table below shows the Froude number for different type of hydraulic jump.

Froude Number (Fr_1)	Type of Hydraulic Jump
($Fr = 1$)	No hydraulic jump (critical flow)

Froude Number ((Fr ₁))	Type of Hydraulic Jump
(1 < Fr ₁ ≤ 1.7)	Undular jump (weak, unstable)
(1.7 < Fr ₁ ≤ 2.5)	Weak jump
(2.5 < Fr ₁ ≤ 4.5)	Oscillating jump
(4.5 < Fr ₁ ≤ 9)	Stable (steady) hydraulic jump
(Fr ₁ > 9)	Strong jump

A stable and well-balanced hydraulic jump is generally obtained when:

$$4.5 < Fr_1 \leq 9.0$$

The importance of water management work in the distribution system of canal are listed below:

- i. Water management work regulates the distribution of water and ensures fair supply to all the farmers from the front end to the tail end.
- ii. It helps to deliver the desirable quantity of water at the required time.
- iii. It helps to avoid over-irrigation that raises the groundwater table.
- iv. It also helps to regulate the discharge in the canal by the use of gates, regulators, outlets and measuring devices.
- v. Water management works ensures reliable irrigation for agricultural production.
- vi. It helps to improve crop yields and farm income.
- vii. It allows the larger command area to be irrigated with the same water supply.

6. What do you mean by water logging of land? What are the factors responsible for infertility of water-logged land? Describe in the brief the anti-water logging measures.

Answer:

Water logging is the condition of saturation of soil such that air phase is restricted and anaerobic conditions prevails. Waterlogging is often accompanied by soil salinity as waterlogged soils prevent leaching of the salts imported by the irrigation water. Due to water logging, groundwater table rises close to the ground surface (generally within 1.5–2.0 m), causing the root zone to remain saturated and reducing soil aeration.

- Water logging is an increasing problem in flat terrains of terai region and low-lying pocket lands in Nepal.
- It is a moderately serious, localized, but recurring seasonal problem.
- It is primarily driven not only by over-irrigation but also by heavy monsoon rainfall combined with unmanaged infrastructure.
- Water-logging is caused by intensive irrigation, seepage from canals/ reservoirs, seepage from adjoining lands, lack of adequate drainage systems, monsoon rainfall and so on.
- Water logging has caused hamper in nitrification which reduces the crop yield and also leads to rise of salt in surface layer and overgrowth of weeds.

The factors responsible for infertility of water-logged land are as follows:

- i. In water-logged land, the water completely fills the pores of the soil mass and prevents air circulation which reduces the root growth and nutrient uptake of the plants.
- ii. It also results in accumulation of harmful salts as high groundwater levels bring the dissolved salts to the surface through capillary action.
- iii. Lack of oxygen in water-logged soils causes anaerobic decomposition which leads to formation and accumulation of toxic substances such as hydrogen sulphide, methane and ferrous compounds.
- iv. Waterlogged soils warm up more slowly than well-drained soils which reduces the soil temperature.
- v. It results in nutrient deficiency and imbalances due to loss of essential nutrients like nitrogen.

The various anti-water logging measures are given below:

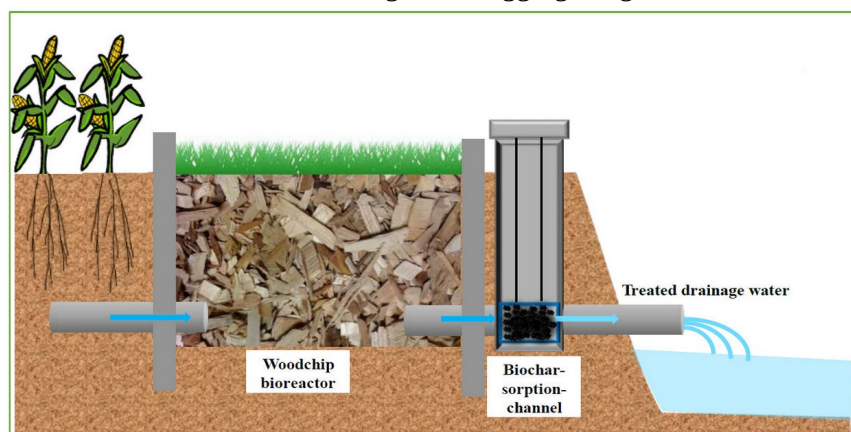
i. Surface drainage system:

- The removal of excess rainwater falling on the fields or excess irrigation water applied to the fields by constructing open ditches. Field drains and other structures is known as surface drainage system.
- The land is loped toward the ditches or drains making excess flow of water to remove from the drains.
- Surface drains can be divided into surface drain and deep surface drains.



ii. Subsurface drainage system:

- The removal of excess rainwater by using buried perforated pipes or tile drains to lower the groundwater table is known as subsurface drainage system.
- It is most effective for controlling water logging in agricultural fields.



iii. Interceptor drains:

- These drains are constructed along the canals or hill slope to intercept seepage before it reaches agricultural land.

iv. Bio-drainage:

- Plantation of deeply rooted trees, highly water absorbing plants to lower the groundwater table naturally.

v. Open drainage system:

- It consists of open ditches that carry water to nearby rivers or canals.
- It is simple and economical for large agricultural areas.

7. What are the steps you consider necessary in the planning and design of Janata ko Bandh? Is it a feasible and sustainable option in mitigating water induced disaster in Nepal?

Answer:

Janata ko Bandh (People's Embankment) is a community-based flood protection embankment constructed mainly using local materials, local labor, and community participation to protect settlements, agricultural land, and infrastructure from river flooding and erosion.

Janata ko Tatbandha program in Nepal was officially started in the FY 2066/2067 BS. It was launched by the Government of Nepal to control soil erosion and river-induced flooding along major rivers, particularly targeting vulnerable settlements in the Terai plains. The effectiveness of this program was mixed, yielding localized successes in flood prevention alongside persistent challenges with implementation delays and disputes.

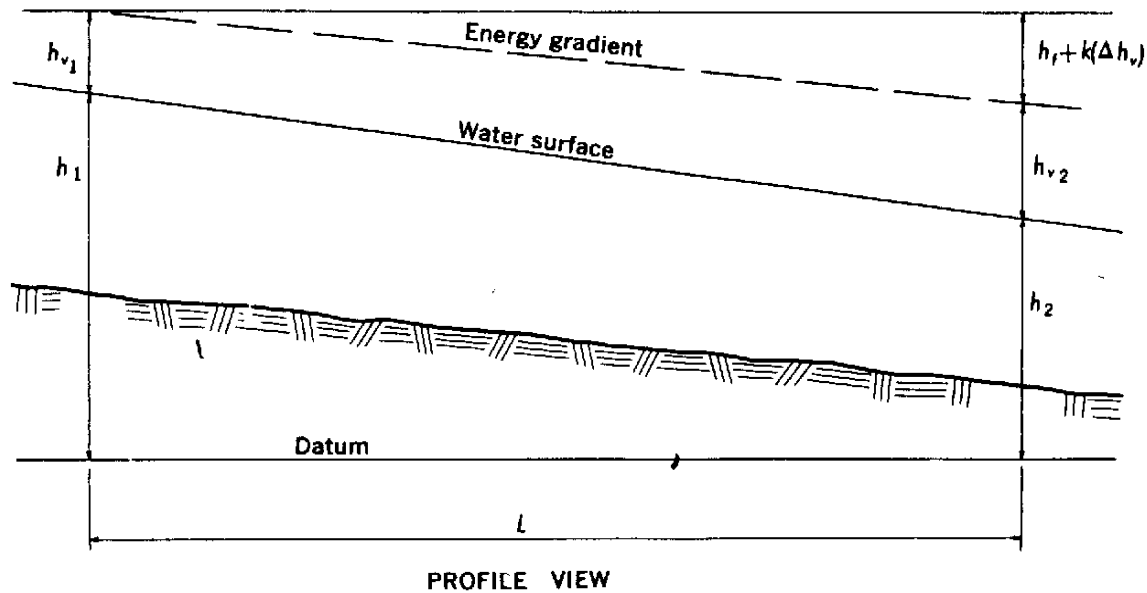
The steps necessary in the planning and design of Janata ko Bandh are as follows:

- i. Reconnaissance survey:
 - Identify the project area and water source.
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 - ii. Feasibility study:
 - Assess water availability (surface or groundwater).
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 - Prepare structural drawings and reinforcement details.
 - vi. Environmental assessment:
 - Conduct IEE/EIA as required.
 - Prepare environmental management and mitigation plans.
 - Address land acquisition, resettlement, and social impacts.
 - Preparation of Detailed Project Report:
 - Prepare design drawings.
 - Estimate quantities and project cost.
 - Prepare construction specifications and implementation schedule.
 - vii. Construction planning and implementation:
 - Procure contractors and materials.
 - Construct headworks, canals, and associated structures.
 - Supervise quality control and project progress.
 - Use of locally available construction materials and labors.
 - viii. Operation and maintenance:
 - Regular inspection especially before and after monsoon seasons.
 - Erosion, settlement and damaged sections are promptly repaired.
 - Local community, user's committee are responsible for operation and maintenance.
8. **Define a point of rainfall. Describe the procedures for slope area method of computing flow discharge in a river.**

Answer:

- The depth of rainfall measured at a single location by a rain gauge during a specified period of time is known as point of rainfall.
- It represents the rainfall at that particular station and is expressed in millimeters (mm).

Slope-area method is an indirect technique to compute peak flood discharge in a river using high-water marks, channel geometry, and Manning's roughness equation.



- Selection of suitable river reach:
 - A straight, stable and uniform reach of river is chosen.
 - The bends, confluences, bridges and backwater effects are avoided.
- Conduction of cross-section survey:
 - At least two cross-sections are surveyed.
 - Flow area and wetter perimeter is measured.
- Identification of high flood marks:
 - The flood marks are located on both river banks.
 - Flood water surface elevations are determined.
- Measurement of reach length:
 - The distance between the selected cross-sections along the river is to be determined.
- Calculation of water surface slope:
 - The difference in flood water levels between the two sections is determined.
 - Slope is computed as:

$$S = \frac{\Delta h}{L}$$

where:

S = Water surface slope

Δh = Difference in water level

L = Length of river reach

vi. Calculation of hydraulic radius:

➤ Compute:

$$R = \frac{A}{P}$$

where :

$$A = \text{Cross-sectional area (m}^2\text{)}$$

$$P = \text{Wetted perimeter (m)}$$

vii. Selection of Manning's coefficient:

➤ The appropriate value of Manning's roughness coefficient is estimated based on the river bed material, vegetation etc.

viii. Calculation of flow velocity:

➤ Flow velocity is computed by using the formula:

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

ix. Computation of river discharge:

➤ The river discharge was determined using:

$$Q = AV$$

where :

$$Q = \text{Discharge (m}^3/\text{s)}$$

$$A = \text{Flow area (m}^2\text{)}$$

$$V = \text{Mean velocity (m/s)}$$

9. Flood flow estimation of 1 in 50 years return period needs to be estimated to design a head works and intake structures. The method used will also depend on the validity of a data at a point of interest. Keeping in mind of above, describe the various methods practiced in Nepal to estimate the flood flows of catchment?

Answer:

A design flood is the flood discharge adopted for the safe design of hydraulic structures such as headworks, spillways, bridges etc. For a 50-year return period, the estimated flood has a 2% probability of being equaled or exceeded in any year.

The different methods of flood flow estimation are as follows:

i. Rational method:

Rational method is suitable for the catchment with limited catchment area)

$$Q = CIA/360$$

where:

Q = Peak runoff (m³/s)

C = Runoff coefficient

I = Rainfall intensity (mm/hr)

A = Catchment area (ha)

ii. **Empirical method:**

Empirical formula is suitable for the catchment with large catchment area.

Dicken's Formula:

$$Q = C A^{3/4}$$

where:

Q = Peak runoff (m^3/s)

A = Catchment area (km^2)

C = Dicken's coefficient (depends on region)

Ryve's Formula:

$$Q = C A^{2/3}$$

where:

Q = Peak runoff (m^3/s)

A = Catchment area (km^2)

C = Ryve's coefficient

iii. **Unit hydrograph method:**

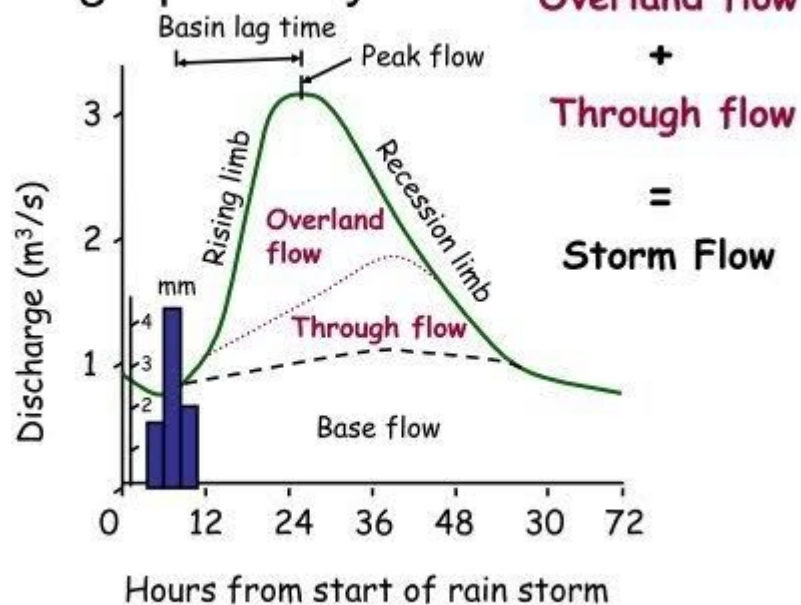
Unit hydrograph is the direct run of hydrograph resulting from unit depth (1cm) of excess rainfall occurring at the uniform rate for the specific duration of D hours.

Unit hydrograph is the unit pulse response function of linear hydrologic system.

The basic assumptions of the unit hydrograph theory in flood forecasting in rivers of Nepal are listed below:

- The excess rainfall has constant intensity one by $\frac{1}{D}$ cm/hr for effective storm duration of D hours.
- The excess rainfall is uniformly distributed over the entire catchment area.
- The base time of direct runoff hydrograph (DRH) resulting from excess rainfall of given duration is constant.

Hydrograph Analysis



10. Distinguish clearly between a shallow well in confined aquifer? Show with the help of sketches the various types of wells?

Answer:

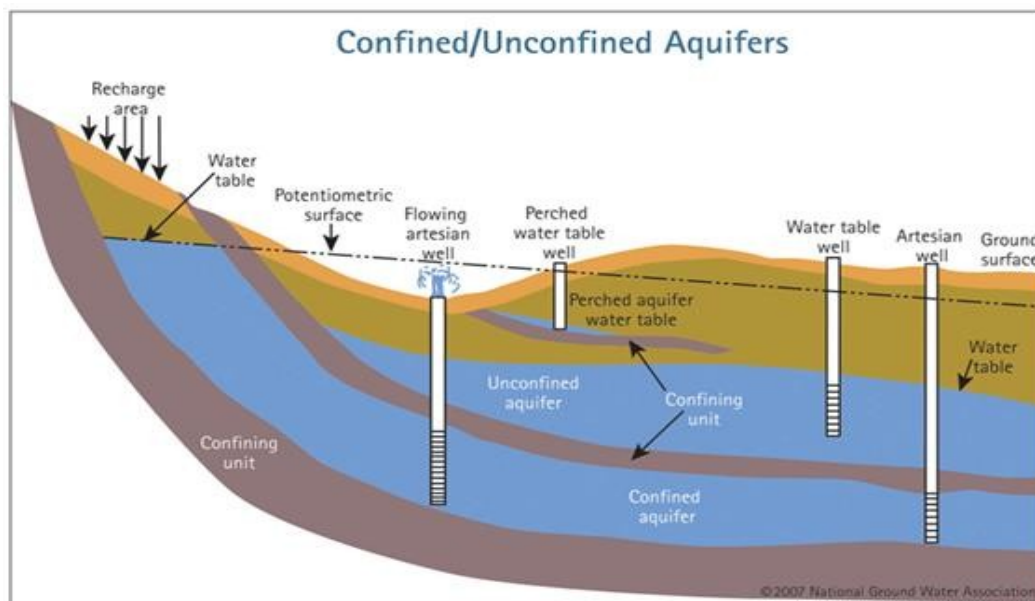
- An aquifer may be defined as the geological formation that contains sufficient permeable material that permits storage as well as transmission of water through it under ordinary conditions.
- Water can be withdrawn from aquifer for example: sand layer.
- Aquifer are of two types:

i. Confined aquifer:

- It is the type of aquifer in which ground water is confined under pressure greater than atmospheric pressure by overlying relatively impermeable strata.
- Confined aquifer generally occurs at significant depth below the ground surface.
- It is known as artesian aquifer or pressure aquifer.

ii. Unconfined aquifer:

- It is the type of aquifer in which water table forms the upper surface of zone of saturation.
- It is also known as phreatic aquifer or non-artesian aquifer.



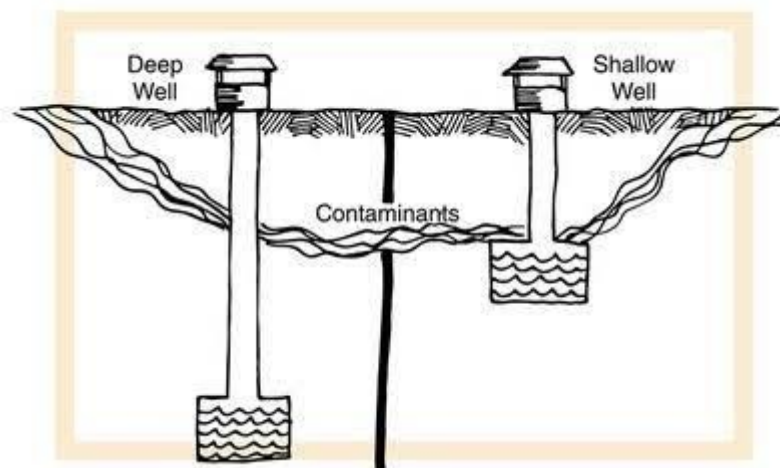
The differences between confined and unconfined aquifer are as follows:

Unconfined aquifer	Confined aquifer
i. There is no confining upper layer in unconfined aquifer	i. There is an impermeable upper layer in confined aquifer.
ii. Water in unconfined aquifer remains at atmospheric pressure.	ii. Water in confined aquifer is under pressure.
iii. Unconfined aquifer is recharged by direct rainfall and infiltration.	iii. Confined aquifer is recharged from the distant exposed areas.
iv. Storage depends on seasonal recharge and may fluctuate significantly.	iv. Storage in confined aquifer is larger and is more reliable water storage.
v. Examples: shallow sand or gravel aquifer near the ground surface.	v. Examples: deep sandstone aquifer beneath clay layers.

Wells:

- Wells are the vertical holes made below the ground surface to tap water.
- The maximum quantity of water that can be guaranteed during lean period is known as the safe yield of the well.
- Water wells may be classified as:
 - i. Open wells or dug wells:**
 - They are comparatively of larger diameter and larger depth but low yields.
 - The diameter ranges from 1 to 10 m usually with the depth at the range of 2 to 20 m.

- The thickness of open wells varies from 50 cm to 75 cm depending on the depth of the well.
- Open wells may further divide into:
 - a. Shallow open wells:
 - ✓ Shallow open wells are those which rest in the top water-bearing strata and draw their supplies from the surroundings.
 - ✓ There is always a risk of contamination because water is obtained from the top most pervious layer.
 - b. Deep open wells:
 - ✓ Deep open wells are those which rest on impervious strata and draw their supplies from pervious formation lying below the impervious strata.
 - ✓ The quantity of water is greater in this well than in shallow wells.
 - ✓ Deep well water has better quality but sometimes it may contain dissolved impurities as it depends upon geographical conditions.



ii. Tube wells:

- Tube well is a long pipe sunk into the ground intercepting one or more water bearing strata.
- As compared to open wells the diameter of tube wells is much less.
- The depth of tube wells up to 30 m is shallow tube wells and up to 600 m are deep tube wells.
- The tube wells may be classified as:
 - a. **Strainer type tube wells:**
 - It is the most common and widely used tube well.
 - The pipe introduced into the ground is an assembly of strainer pipes and blind pipes that are alternately placed.
 - The length of strainer and blind pipes are so adjusted that the strainer pipe rest against the water bearing strata and blind pipes rest against impervious strata.
 - b. **Cavity type tube wells:**
 - A cavity-type tube wells consists of a pipe sunk into the ground and resting on the bottom of strong strata.
 - The initial stage of pumping fine sand comes out with water consequently, a cavity is formed at the bottom.
 - c. **Slotted type tube wells:**
 - It consists of a pipe that is slotted for part of its length at one end and the rest of the length is a plain pipe.

- The mixture of gravel and coarse sand is placed around the pipe to prevent the fine sand from entering the well pipe.
- The slotted portion of the pipe is surrounded by a mixture of gravel and coarse sand called shrouding which is filled between casing pipe and slotted pipe during withdraw of casing.

d. Perforated type tube wells:

- It is a shallow depth tube well and is used for a short duration such as in construction site.
- Pipes having covered by jute ropes act as a strainer and prevents the fine particles from entering the well pipe.

The different types of wells suitable for different types of aquifers are listed below:

i. Open well:

- Suitable for unconfined (shallow) aquifers.
- Large diameter (1–5 m) and shallow depth.

ii. Tube well:

- Suitable for confined and unconfined aquifers.
- Small diameter and deep.
- Constructed by drilling and fitted with casing and screens.

iii. Artesian well:

- Suitable for confined aquifers.
- Water rises naturally due to artesian pressure.

iv. Driven well:

- Suitable for shallow unconfined aquifers in sandy or gravelly soils.
- Constructed by driving a small-diameter pipe into the ground.

v. Bored well:

- Suitable for soft formations and shallow unconfined aquifers.
- Constructed using hand or power augers.

vi. Radial well:

- Suitable for alluvial aquifers near rivers.
- Consists of a central well with horizontal perforated pipes extending into the aquifer.